

AI & ML Notes - Beginner to Intermediate

1. Introduction to AI

Artificial Intelligence (AI) is the simulation of human intelligence in machines. It includes learning, reasoning, and self-correction.

2. Machine Learning Overview

Machine Learning (ML) is a subset of AI that enables systems to learn patterns from data without being explicitly programmed.

3. Types of Machine Learning

Supervised Learning: Uses labeled data.

Unsupervised Learning: Finds hidden patterns.

Reinforcement Learning: Learns via rewards and penalties.

4. Key Algorithms

Linear Regression, Logistic Regression, Decision Trees, Random Forest, KNN, SVM, Neural Networks.

5. Deep Learning

Deep Learning uses neural networks with multiple layers to learn complex patterns in data like images, speech, and text.

6. Applications

AI/ML is used in healthcare, finance, recommendation systems, autonomous vehicles, and natural language processing.

7. Data Preprocessing

Includes cleaning data, handling missing values, normalization, feature selection, and encoding categorical variables.

8. Model Evaluation

Accuracy, Precision, Recall, F1 Score, Confusion Matrix, ROC-AUC are common evaluation metrics.

9. Overfitting & Underfitting

Overfitting: Model learns noise.

Underfitting: Model is too simple to learn patterns.

10. Future of AI

AI is rapidly evolving with generative models, transformers, and large language models shaping the future.